

Lesson 6

Topic: Creating Basic Visualizations & Adding Interactivity to Visuals

1. Name three types of visuals you can create in Power BI

- **Bar/Column Chart** (comparisons across categories)
 - **Line Chart** (trends over time)
 - **Pie/Donut Chart** (proportional data)
- (Other examples: Map, Card, Matrix, Funnel, Gauge, Waterfall, etc.)
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2. How do you add a slicer to a report?

- In **Report view**, go to the **Visualizations pane** → click **Slicer** icon.
 - Drag a field (e.g., *Quarter* or *Region*) to the slicer's **Values**.
 - Adjust orientation (dropdown/list/tiles) in the **Format pane**.
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3. Difference between a bar chart and a column chart

- **Bar chart** → categories on the **Y-axis**, values on the **X-axis** (horizontal bars).
 - **Column chart** → categories on the **X-axis**, values on the **Y-axis** (vertical bars).
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4. How do you change the color of a visual background?

- Select the visual → **Format pane** (paint roller icon) → **Background** → turn it **On** → choose a color and adjust transparency.
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5. What does "drill-down" mean in a visual?

- It lets users **navigate hierarchical data levels**.
Example: Click on *Region* bar → drill into *Product* → drill further into *Quarter*.
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6. Create a bar chart showing SalesAmount by Region

- Insert a **Clustered Bar Chart**.
 - Put **Region** in *Axis*.
 - Put **SalesAmount** in *Values*.
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7. Add a slicer for Quarter to filter all visuals on the page

- Insert **Slicer visual**.
 - Add **Quarter** field.
 - Make sure slicer interactions apply to all visuals (default behavior unless edited).
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8. Format the bar chart to show data labels

- Select the bar chart → **Format pane** → turn **Data labels** = **On**.
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9. Use a line chart to show SalesAmount trends over Quarter

- Insert a **Line Chart**.
- Add **Quarter** to *Axis*.

- Add **SalesAmount** to *Values*.

10. Add a tooltip to display Product details when hovering over bars

- Create a **Tooltip page** (Page Information → set as Tooltip).
- Add **Product details visual(s)** there.
- Assign the Tooltip page in the bar chart → **Format pane** → **Tooltip** → **Report page**.

11. Sync slicers across multiple report pages

- Select a slicer → **View ribbon** → **Sync slicers**.
- Choose which pages to apply and whether the slicer is visible or hidden on each.

12. Create a custom visual with dynamic measure selection (Sales vs. Profit)

- Create a **disconnected table** with values: *Sales, Profit*.
- Create a **measure** like:
 - Selected Measure =
 - SWITCH(
 - SELECTEDVALUE(MeasureTable[Measure]),
 - "Sales", SUM(Sales[SalesAmount]),
 - "Profit", SUM(Sales[ProfitAmount])
 -)
- Use this measure in visuals, and add the slicer from MeasureTable.

13. Implement a hierarchy for Region > Product > Quarter drill-down

- In **Fields pane**, create a hierarchy by dragging fields: **Region** → **Product** → **Quarter**.
- Add hierarchy to Axis → enable drill-down arrows in the visual.

14. Use bookmarks to toggle between two visuals in the same space

- Place two visuals in the same location.
- Create **Bookmark 1** (only first visual visible).
- Create **Bookmark 2** (only second visual visible).
- Add a **Button/Shape** to switch between bookmarks.

15. Optimize a slow-rendering report with 10+ visuals

- **Remove unnecessary visuals** or split across pages.
- **Reduce data size** (aggregations, fewer rows/columns).
- **Turn off auto date/time**.
- **Limit high-cardinality slicers**.
- **Disable unnecessary interactions**.
- **Use measures instead of calculated columns** where possible.
- **Optimize DAX** (remove iterators, use SUMX only if needed).